Introduction to Internet of Things

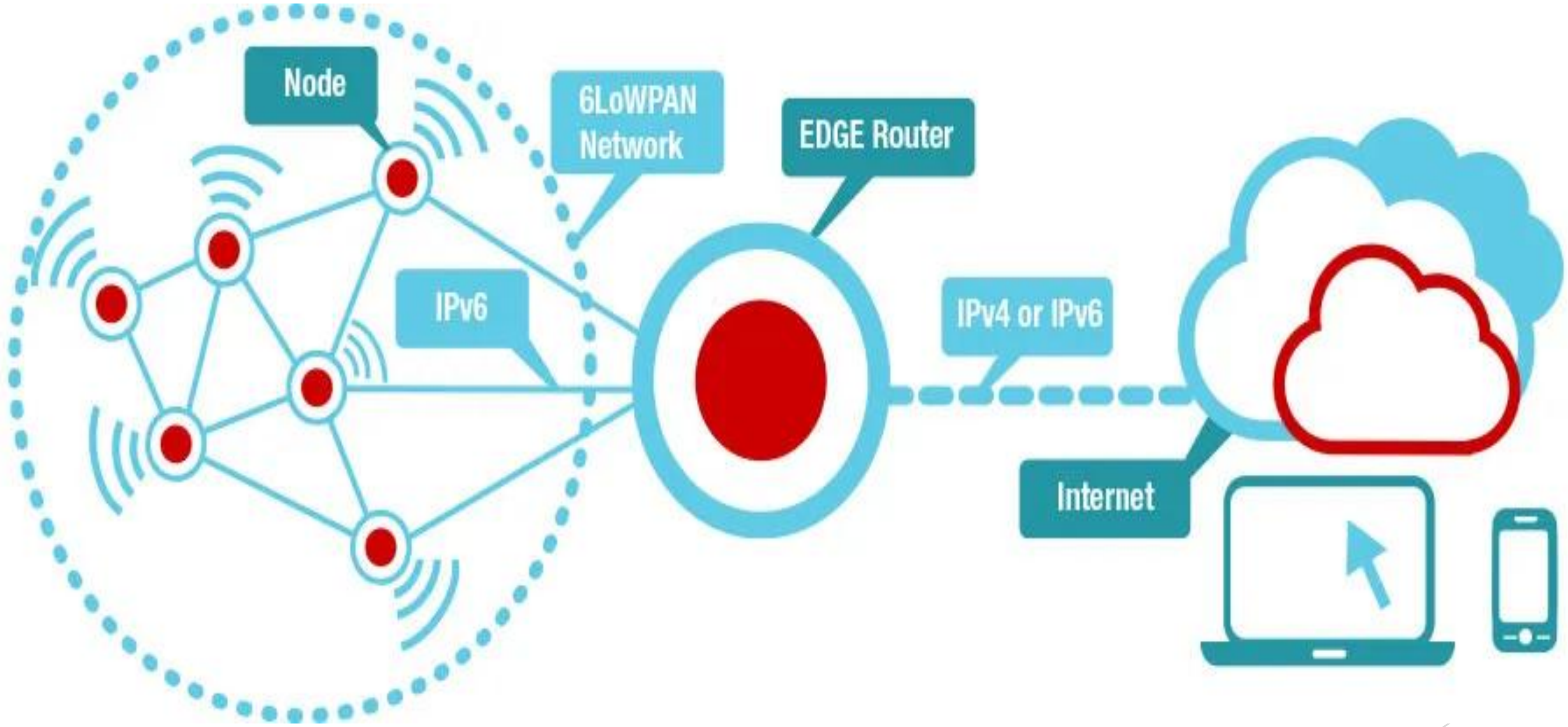
Lecture 22&23

6LoWPAN

- 6LoWPAN abbreviates IPv6 over Low Power Wireless Personal Area Networks.
- It is a lightweight adaptation layer that allows IPv6 packets to be transmitted over low-power wireless networks, such as IEEE 802.15.4.
- An adaptation layer modifies or translates data, protocols, or signals so that two incompatible layers or systems can communicate effectively.
- A simple IoT network uses direct device-to-device communication without IP, while an advanced IoT network uses IP-based communication with routing and Internet connectivity.

Compresses IPv6 headers so they fit in small 802.15.4 frames

6LoWPAN Architecture



6LoWPAN and its functions

- Inside the mesh network, routing is handled by RPL, which build a DODAG graph.

Functions:

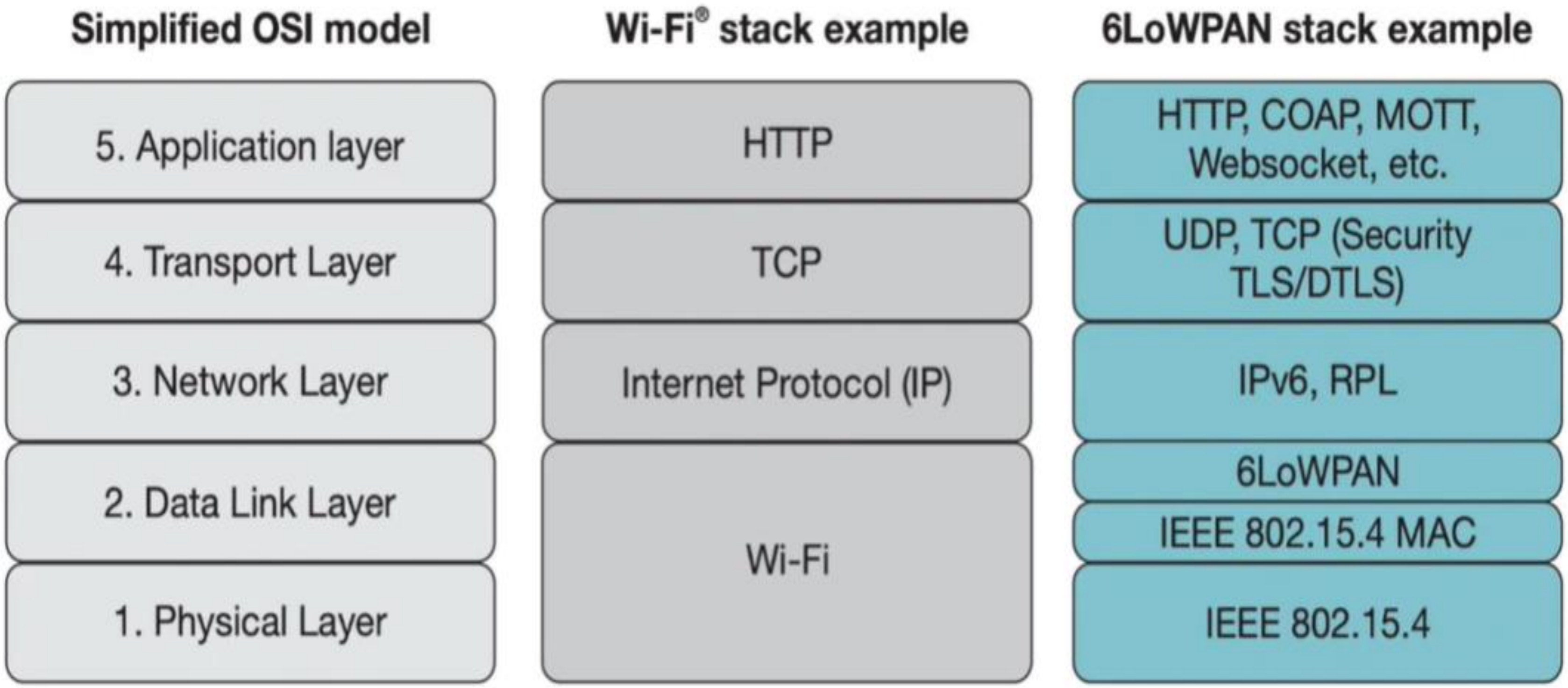
1. IPv6 header compression
2. Fragmentation and reassembly
3. Adaptation Between IPv6 and 802.15.4
4. Address Mapping
5. Support for Mesh and Star Networks
6. Low Power Optimization

It allows for the smallest devices with limited processing ability to transmit information wirelessly using an IP

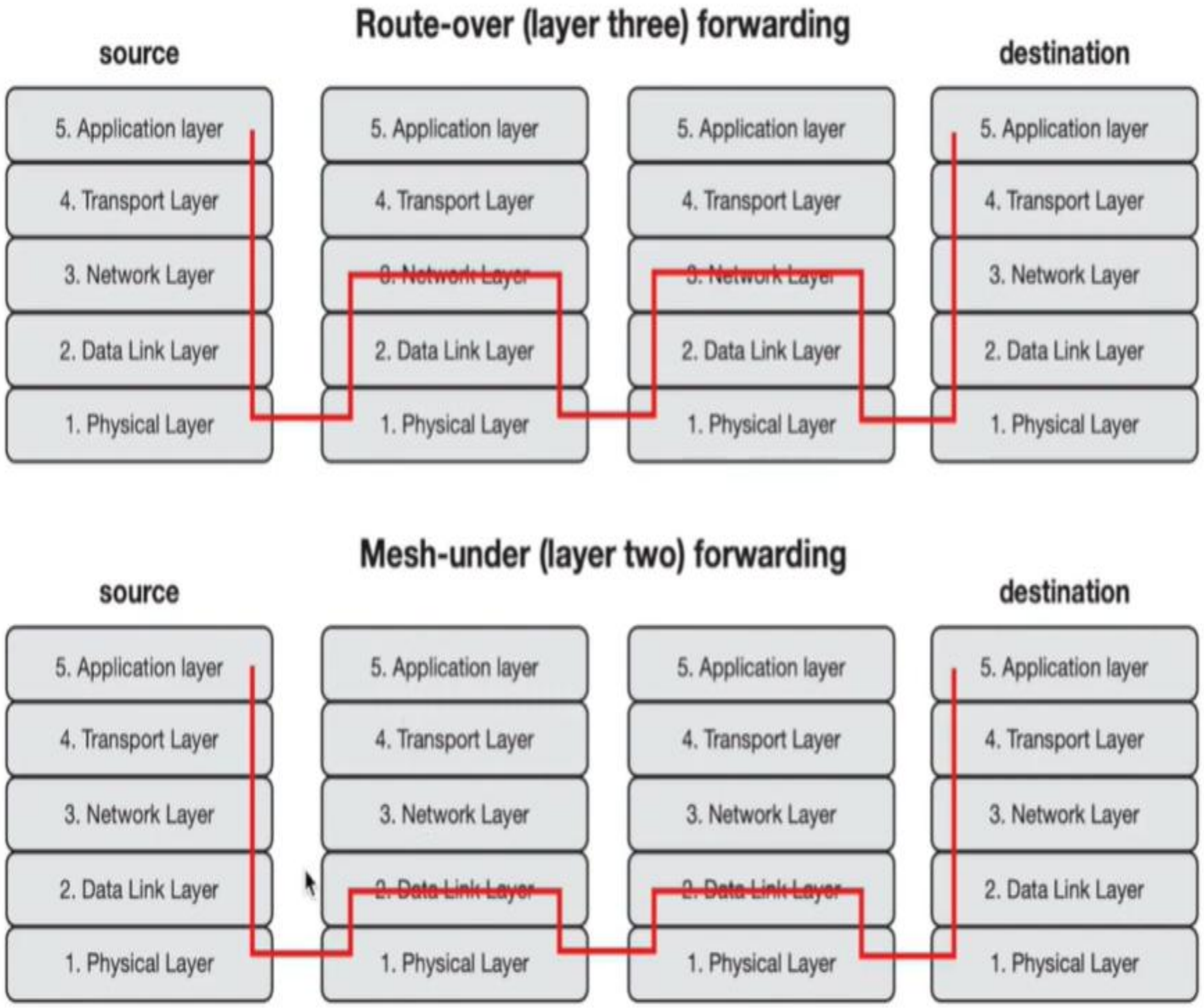
Features

1. **Header Compression:** 6LoWPAN uses header compression techniques to reduce the size of IPv6 packets, making them more efficient to transmit over low-bandwidth wireless networks.
2. **Fragmentation:** 6LoWPAN can fragment large IPv6 packets into smaller packets that can be transmitted over the low-bandwidth network.
3. **Addressing:** 6LoWPAN uses 16-bit addresses instead of the 128-bit addresses used in IPv6, to reduce the size of the packet headers and to conserve battery life in low-power devices.
4. **Mesh Networking:** 6LoWPAN supports mesh networking, by which devices can forward packets for each other to extend the range of the network.
5. **Low Power Consumption:** 6LoWPAN is designed to be energy-efficient, with low overheads for packet transmission and minimal power requirements for devices.
6. **Security:** 6LoWPAN includes support for security mechanisms such as encryption and authentication to protect the privacy and integrity of data transmitted over the network.

6LoWPAN stack:



6LoWPAN adaptation layer (Routing):

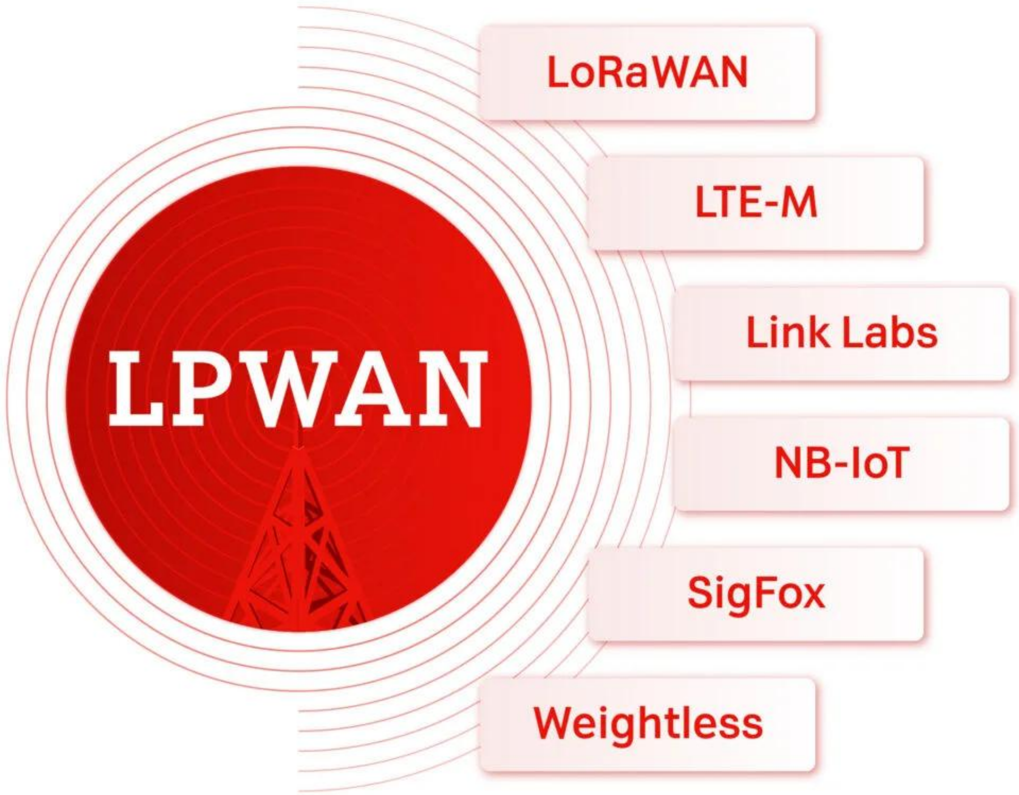


LPWAN

LPWAN

- It is a wireless technology designed for IoT, providing long-range communication (kilometers) and years of battery life for devices transmitting small data volumes.
- Typical characteristics of LPWAN are low operating costs and a long range, making it ideal for IoT applications.
- Devices connected via LPWAN also have a long battery life and LPWAN technologies such as LTE-M and NB-IoT offer reliable connections, particularly useful for applications such as smart cities, industrial automation and environmental monitoring.

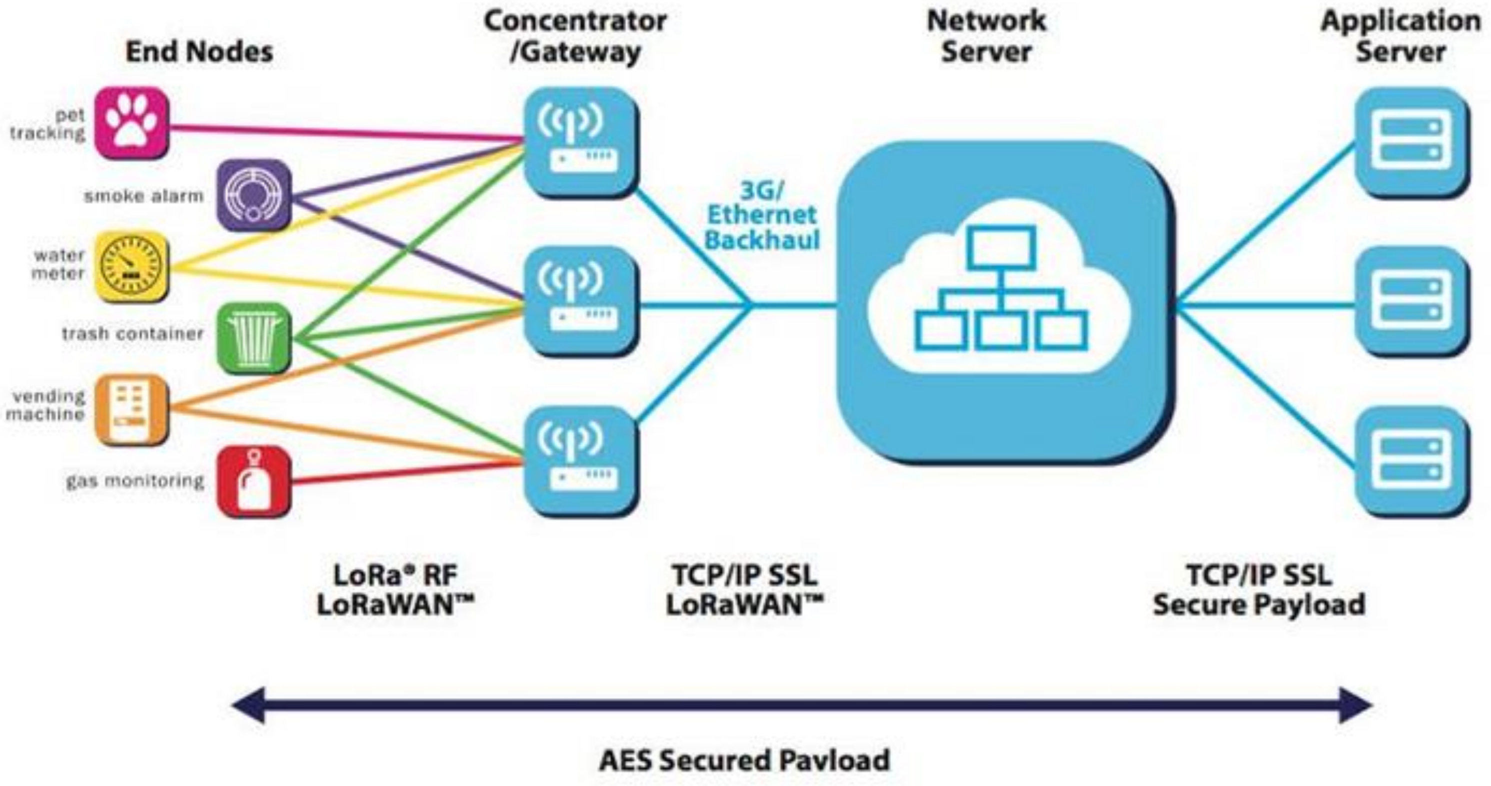
The network technology LPWAN (Low Power Wide Area Network) was developed for the needs of the Internet of Things (IoT) as it enables devices to be connected over long distances with minimal energy consumption.



LoRaWAN: Long range and low energy consumption

- LoRaWAN stands for Long Range Wide Area Network.
- This technology is characterized by its long range and low energy consumption, ideal for applications that need to transmit data over long distances.
- LoRaWAN uses the license-free spectrum (like ISM bands), which reduces operating costs.

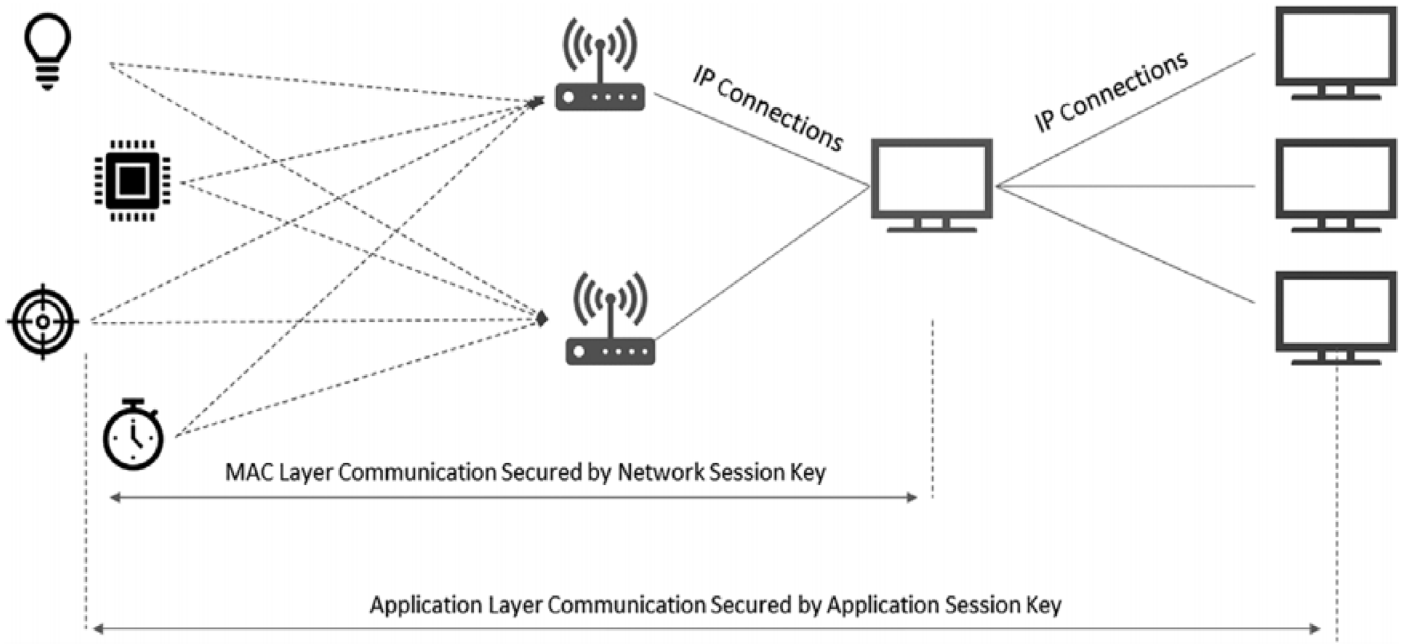
LoRaWAN



End Nodes
Concentrator /Gateway
Network Server
Application Server
pet tracking
smoke alarm
water meter
trash container
vending machine
gas monitoring
3G/ Ethernet Backhaul
LoRa® RF LoRaWAN™
TCP/IP SSL LoRaWAN™
TCP/IP SSL Secure Payload
AES Secured Payload

LoRa network Architecture

- The networks do only support low data transfer rates.
- LPWAN networks can also connect numerous devices simultaneously, making them ideal for large-scale IoT implementations.



LoRa

- LoRaWAN specifies the network's communication protocol and system architecture, while the LoRa physical layer allows long distance communication links.
- LoRa is a radio signal transmission techinque that uses chirped, multi-symbol data to communicate information.

LoRa

- **Phy**sical Layer
- Modulation Technique
- Based on Chirp Spread Spectrum
- Point-to-Point communication

LoRaWAN

- **MAC** Layer (Protocol)
- System Architecture
- Defines message format & security
- Manages the network

Key Features of LoRa

- **Long Range:** LoRa can reach distances of up to 10 kilometers in rural areas and up to 2 kilometers in urban areas.
- **Low Power:** LoRa devices can operate for years on a single battery charge, making them ideal for IoT applications that require long-term monitoring and sensing.
- **Scalability:** LoRa networks can be deployed in a variety of configurations, from point-to-point setups to large-scale, wide-area networks.
- **Security:** supports various security mechanisms such as encryption and authentication.
- **Cost Effectiveness** and **Standardization**

Components in LoRa Technology

End Devices: These are IoT devices equipped with LoRa transceivers that transmit data to gateways. They can also be configured to operate in various modes, such as class A, class B, or class C, depending on their power requirements and communication needs.

Gateways: These are devices that receive data from end devices and forward it to a network server using the LoRaWAN protocol. Gateways typically use multiple channels to receive data from different end devices and transmit the data to the network server over a backhaul network such as Ethernet, Wi-Fi, or cellular.

Network Server: The LoRaWAN network server manages communication between the end devices and the gateways. It receives data from gateways and routes it to the appropriate end device using the unique device identifier (DevEUI). The network server also manages the security aspects of the communication, such as authentication and encryption.

LoRaWAN Devices

LoRaWAN class A Devices:

- It has the longest battery life.
- It stays in sleep mode for the majority of the time.
- It can receive messages (downlink) only if it has sent messages (uplink).
- Examples: Fire Alarm, Flood Detector etc.

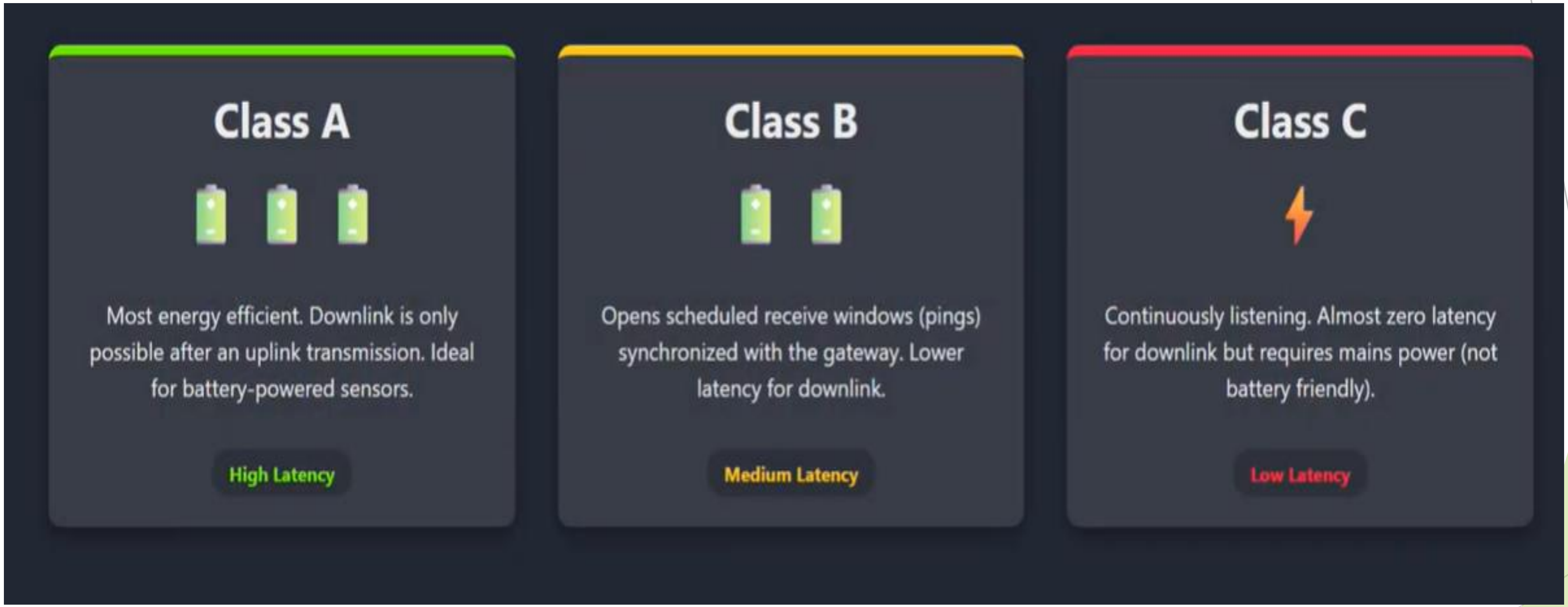
LoRaWAN class B Devices:

- It has an average batery life.
- It listens to the netwok periodically.
- It can receive messages (downlink) even if it has not sent messages.
- Examples: Metering of Temperature, Hunmidity etc.

LoRaWAN Devices

LoRaWAN class C Devices:

- It has the lowest battery life.
- It listens to the network continously.
- It has minimum latency for messages.
- Examples: Traffic Monitoring, Real-time Applications etc.


Class A
Most energy efficient. Downlink is only possible after an uplink transmission. Ideal for battery-powered sensors.
High Latency
Class B
Opens scheduled receive windows (pings) synchronized with the gateway. Lower latency for downlink.
Medium Latency
Class C
Continuously listening. Almost zero latency for downlink but requires mains power (not battery friendly).
Low Latency

NB-IoT

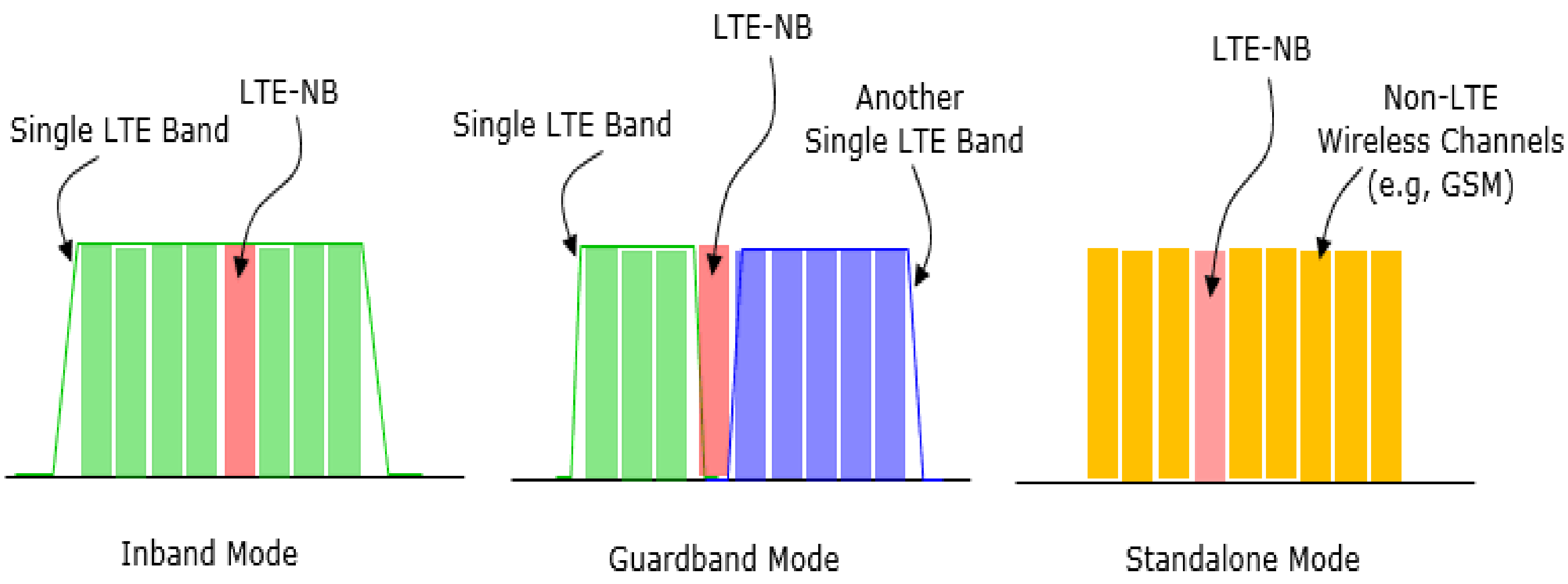
- This network was designed to provide stable and reliable connectivity for IoT devices.
- The concept of NB-IoT is based on the use of cellular networks to provide connectivity to IoT devices.
- NB-IoT networks operate in the licensed spectrum, which ensures a high-quality service and data security.
- NB-IoT uses narrowband radio-frequency channels to transmit data with low power consumption, making it ideal for IoT devices.
- NB-IoT networks have three deployment options: in-band, standalone, and guard-band.
- Use GSM or unused LTE channels.

NB-IoT

In-band operation: It involves using a portion of an existing LTE cellular network to provide NB-IoT coverage, which allows seamless integration of NB-IoT devices into the existing LTE infrastructure.

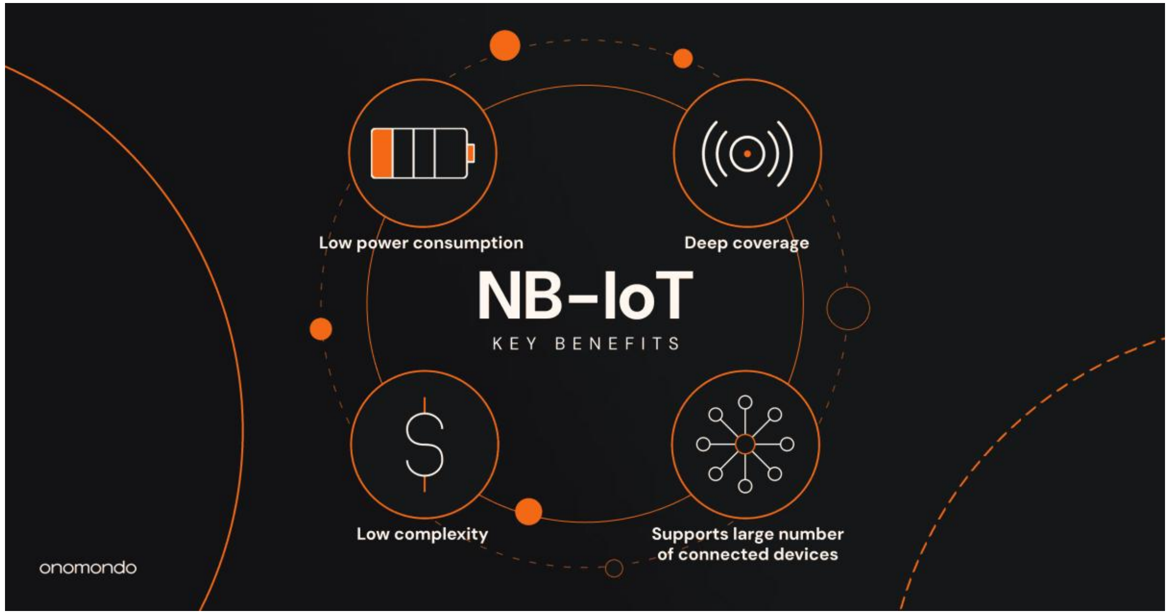
Guard-band operation: It involves using the space between existing cellular networks to deploy NB-IoT, which is an efficient way to utilize scarce resources.

Standalone operation: It is a separate dedicated network built specifically for IoT devices, with no integration with the existing cellular network.


Single LTE Band
LTE-NB
Inband Mode
LTE-NB
Single LTE Band
Another
Single LTE Band
Guardband Mode
LTE-NB
Non-LTE
Wireless Channels
(e.g, GSM)
Standalone Mode

NB-IoT

- This technology is designed to operate in areas with poor coverage, such as underground parking lots, basements, and remote areas.
- NB-IoT also has a coverage range of up to 100 kilometers, which is significantly higher than other LPWAN technologies currently available.



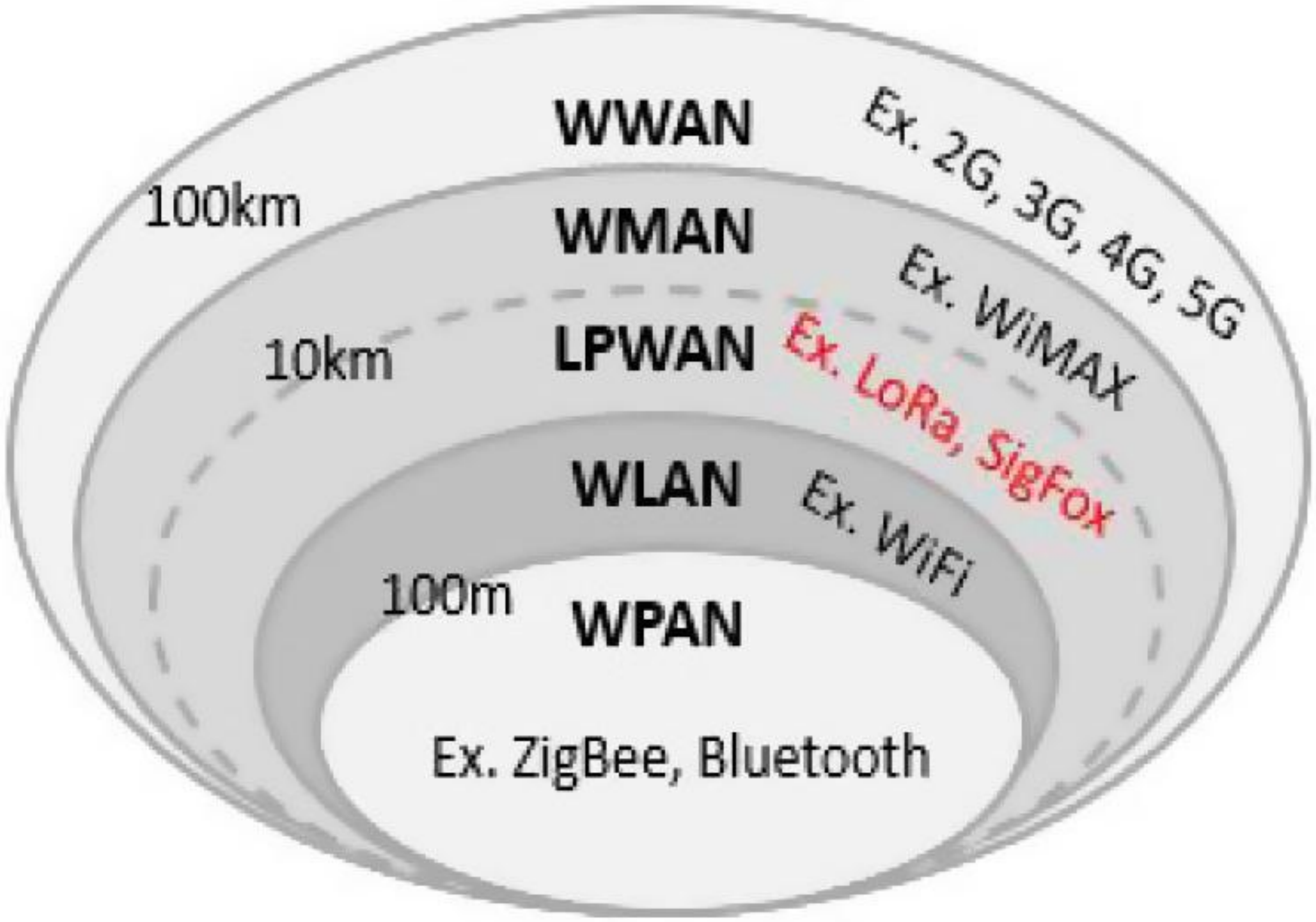
Advantages of NB-IoT

- **Wide Coverage:** NB-IoT provides excellent coverage, even in hard-to-reach areas such as basements, underground spaces, and remote locations.
- **Low Power Consumption:** NB-IoT is designed to operate with low power consumption, enabling battery-powered IoT devices to operate for extended periods without frequent recharging or battery replacement.
- **Scalability:** It can handle a high density of connected devices in a given area.
- **Secure Communication:** NB-IoT incorporates robust security mechanisms to protect IoT data and ensure secure communication.

- **High Reliability:** It is designed to operate in licensed spectrum bands, ensuring interference-free communication. This reliability is particularly important for critical applications such as healthcare monitoring, industrial automation, and public safety systems.
- **Cost Effective:** NB-IoT offers cost advantages in terms of both infrastructure deployment and device connectivity. It utilizes existing cellular infrastructure, allowing for cost-efficient deployment. Additionally, NB-IoT chipsets and modules are designed to be cost effective, making them suitable for low-cost IoT device deployments.
- **Coexistence with Cellular Networks:** NB-IoT can coexist with existing cellular networks, which allows seamless integration and backward compatibility.

LoRa vs NB-IoT

LoRa	NB-IoT
Older (2015)	Newer (2017)
Unlicensed spectrum	Licensed frequency bands
Lower cost per device (but gateway needed)	Higher cost per device (but no gateway needed)
Longer battery life	Shorter battery life
High latency / less frequent data transfer	Low latency / more frequent data transfer
7 to 10 mile range	11 to 13 mile range
Better rural/remote performance	Better urban/dense performance
Lower data rates	Higher data rates (10x LoRa's rates)


WWAN
Ex. 2G, 3G, 4G, 5G
100km
WMAN
Ex. WiMAX
10km
LPWAN
Ex. LoRa, SigFox
WLAN
Ex. WiFi
100m
WPAN
Ex. ZigBee, Bluetooth